

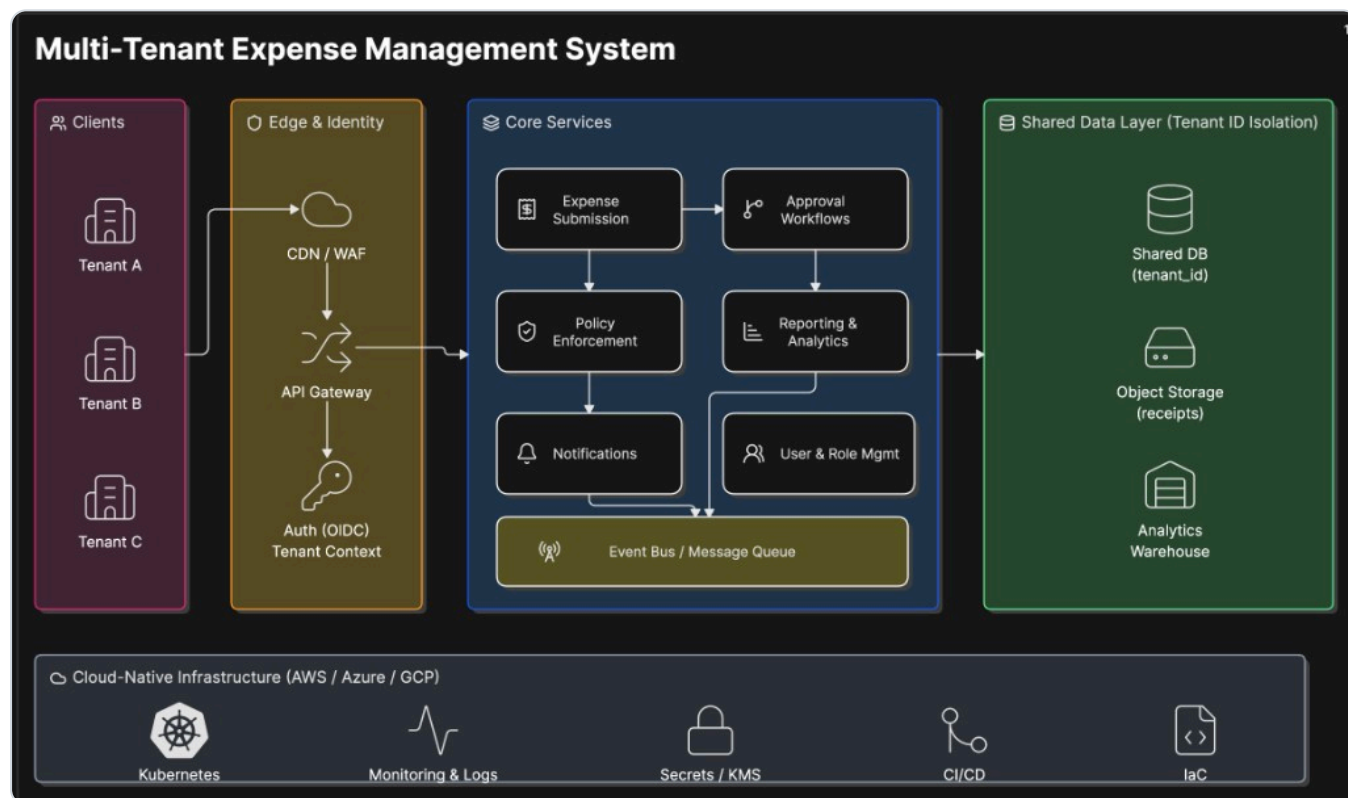
EMS — Technical Documentation (API Reference & UI Walkthrough)

This document describes the implemented Expense Management System: every API endpoint with request/response samples, with **UI screenshots shown inline next to the flow they belong to**.

- **Base URL:** `http://localhost:4000`
- **Format:** JSON over HTTP
- **Source design:** [level-2-design.md](#) · scope: [level-1.5-scope.md](#)

Architecture

Target / production architecture — multi-tenant clients behind an edge + API gateway, stateless core services on a shared data layer with `tenant_id` isolation, on cloud-native infra:



The as-built prototype implements the same core capabilities as a **modular monolith** (single Express app + PostgreSQL + optional Redis + a pluggable blob store for bill uploads, local-disk today); the gateway, event bus, managed object storage (S3), analytics warehouse and infra layers are the documented production path.

1. Conventions

Authentication

- Obtain a token via `POST /auth/login` or `POST /auth/signup` .
- Send it on every protected request: `Authorization: Bearer <accessToken>` .
- The token encodes `{ id, org_id, role, email }` . The server derives `org_id` from the token — clients never pass it. This is the tenant-isolation guardrail.

Roles & permissions

Permission	employee	manager	finance	admin
expense:create / read own	✓	✓	✓	✓
expense:read reportees	—	✓	—	✓
expense:read all	—	—	✓	✓
expense:approve	—	✓	✓	✓
policy:manage / budget:manage	—	—	✓	✓
user:manage	—	—	—	✓
analytics:view	—	✓	✓	✓

Headers

Header	When	Purpose
<code>Authorization: Bearer <token></code>	all protected routes	authentication
<code>Content-Type: application/json</code>	requests with a body	parsing
<code>Idempotency-Key: <unique></code>	approve/reject (optional)	prevents double-apply on retry

Error envelope

All errors share one shape:

```
{ "error": { "code": "FORBIDDEN", "message": "You do not have permission...", "details": null } }
```

HTTP	code	Meaning
400	BAD_REQUEST	Malformed/invalid payload
401	UNAUTHORIZED	Missing/invalid token
403	FORBIDDEN	Authenticated but not allowed
404	NOT_FOUND	Resource not in your tenant
409	CONFLICT	Invalid state transition / duplicate
422	UNPROCESSABLE	Valid shape but business rule failed (budget, currency, no policy)

Notes on data types

- Postgres `NUMERIC` columns are returned as **strings** (e.g. `"3000.00"`), preserving precision. Clients should parse as needed.
- Pagination: list endpoints accept `?limit=` (max 100) and `?offset=` .

2. Auth & Onboarding

POST /auth/signup — create a new organization (tenant onboarding)

Atomically creates an organization, its first **admin** user, a default approval policy and an org budget, then returns tokens (auto-login). Public.

Request

```
{
  "orgName": "Initech",
  "baseCurrency": "INR",
  "adminName": "Bill Lumbergh",
  "email": "admin@initech.test",
  "password": "password123"
}
```

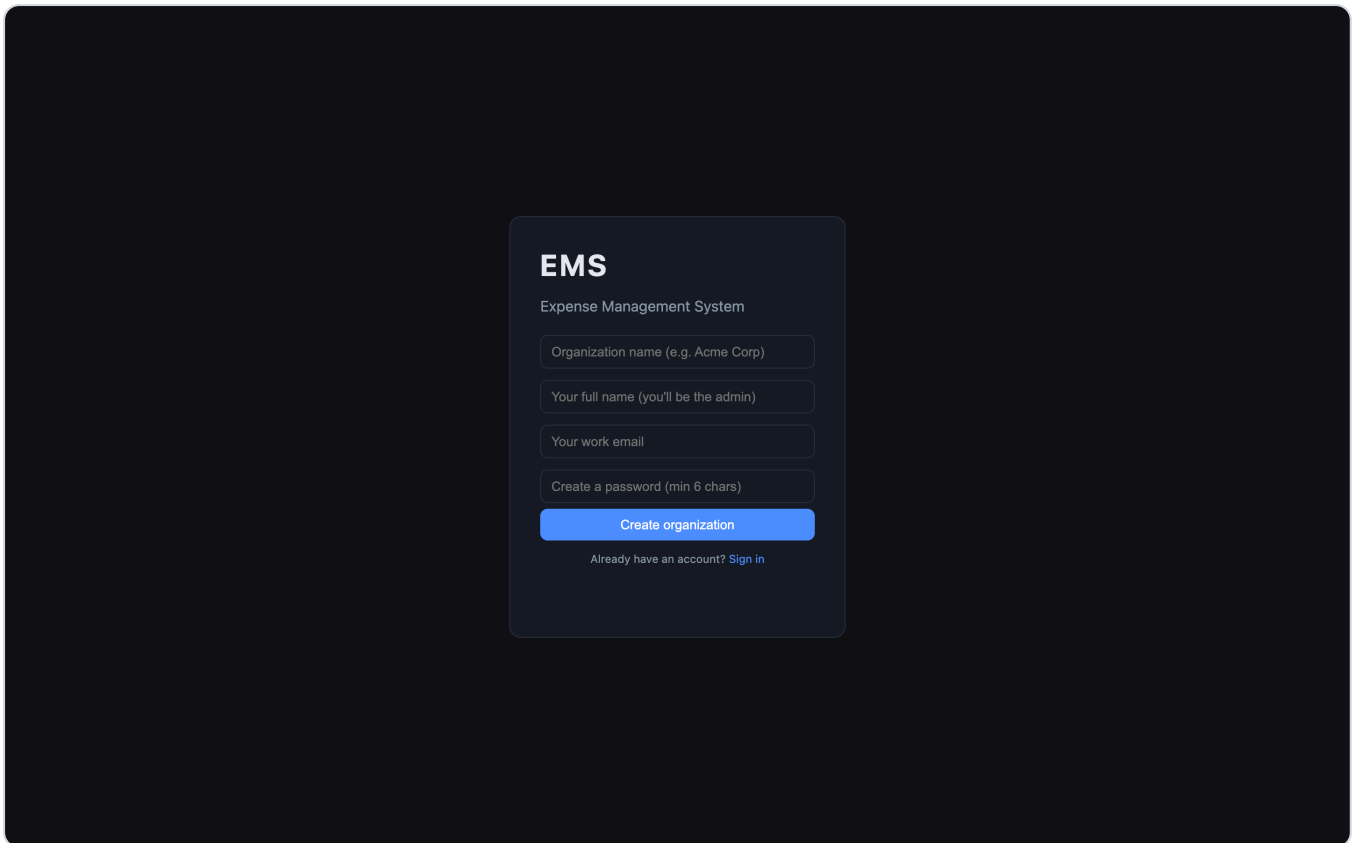
Field	Type	Required	Notes
orgName	string	yes	organization name
baseCurrency	string(3)	no	defaults <code>INR</code>
adminName	string	yes	first admin's name
email	string	yes	unique across all tenants
password	string	yes	min 6 chars

Response 201

```
{
  "accessToken": "eyJhbGciOi...",
  "refreshToken": "eyJhbGciOi...",
  "user": { "id": "uuid", "org_id": "uuid", "name": "Bill Lumbergh", "email": "admin@initech.test",
  "organization": { "id": "uuid", "name": "Initech", "base_currency": "INR" }
}
```

Errors: `400` invalid payload · `409` email already exists.

UI — Sign up (new organization): "Create an organization" provisions a new tenant + its first admin in one step, with a default policy so it's usable immediately.



POST /auth/login

Request

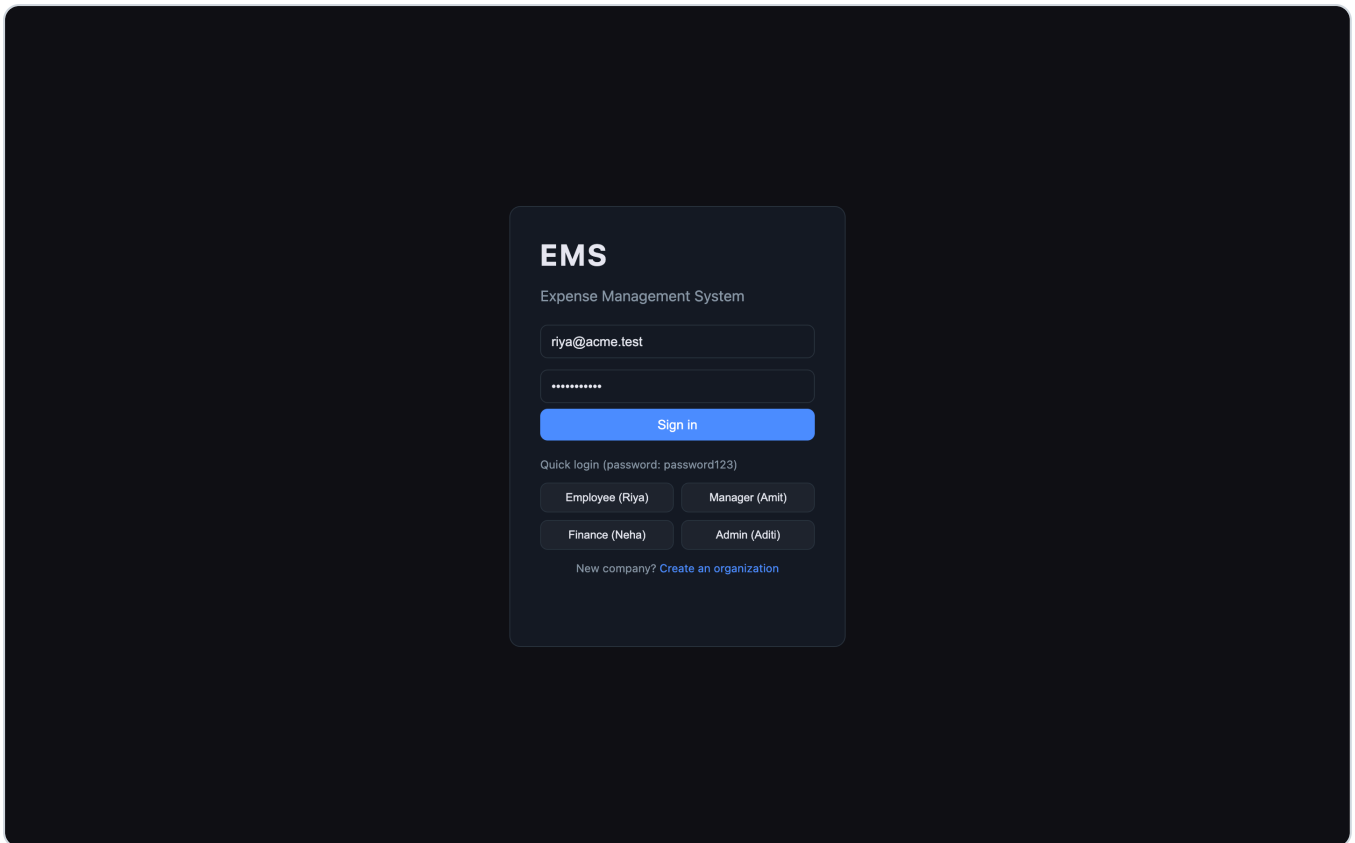
```
{ "email": "riya@acme.test", "password": "password123" }
```

Response 200

```
{
  "accessToken": "eyJhbGciOi...",
  "refreshToken": "eyJhbGciOi...",
  "user": {
    "id": "uuid", "org_id": "uuid", "name": "Riya Sharma",
    "email": "riya@acme.test", "role": "employee",
    "manager_id": "uuid", "is_active": true, "created_at": "2026-..."
  }
}
```

Errors: 401 invalid email or password.

UI — Login: sign in with email/password, or one-click **Quick login** for the seeded roles.



POST /auth/refresh — rotate refresh token

Validates an opaque refresh token, **revokes it (single-use rotation)** and issues a fresh access + refresh pair. Replaying a revoked token is treated as reuse and revokes all of the user's sessions.

Request

```
{ "refreshToken": "<opaque token from login/signup>" }
```

Response 200

```
{ "accessToken": "eyJhbGciOi...", "refreshToken": "<new opaque token>", "user": { "id": "uuid", "...":
```

Errors: 401 missing / invalid / expired / reused token.

POST /auth/logout

Revokes a refresh token. Idempotent. **Request:** { "refreshToken": "<token>" } · **Response 200** → { "revoked": true }

3. System

GET /health

Response 200 → { "status": "ok" }

GET /metrics

Prometheus text exposition. Public.

```
ems_requests_total 128
ems_errors_total 0
ems_request_latency_ms{quantile="0.5"} 4
ems_request_latency_ms{quantile="0.95"} 21
ems_request_latency_ms{quantile="0.99"} 39
```

4. Expenses

POST /expenses — create a draft

Permission: `expense:create` .

Request

```
{ "type": "reimbursement", "category": "travel", "description": "Client visit", "amount": 22000, "cu
```

Field	Type	Required	Notes
type	enum	yes	<code>reimbursement</code> <code>company_paid</code>
category	string	no	defaults <code>general</code>
description	string	no	
amount	number	yes	> 0, in <code>currency</code>
currency	string	no	defaults <code>INR</code> ; supported: INR, USD, EUR, GBP, AED, SGD

The `category` values shown in the UI come from `GET /categories` (admin-managed per org). An expense **cannot be created until the org has an active approval policy** — otherwise there is nothing to route it through.

Response 201

```
{
  "id": "uuid", "org_id": "uuid", "requester_id": "uuid",
  "type": "reimbursement", "category": "travel", "description": "Client visit",
  "amount": "22000.00", "currency": "INR", "base_amount": "22000.00", "fx_rate": "1.000000",
  "status": "draft", "policy_snapshot_json": null, "current_level": 0,
  "sla_due_at": null, "created_at": "2026-...", "updated_at": "2026-..."
}
```

Errors: 400 invalid · 422 unsupported currency · 422 no active policy configured for the org.

GET /expenses — list

Query: `scope=mine|reportees|all` (default `mine`), `status`, `type`, `limit`, `offset` .

- `reportees` requires `expense:read:reportees` ; `all` requires `expense:read:all` .

Response 200 → array of expense objects (newest first). **Errors:** 403 if scope exceeds your permission.

UI — Employee "My Expenses": create reimbursement or company-paid requests and track them with status pills. Foreign-currency amounts show the base (₹) conversion inline.

TYPE	CATEGORY	AMOUNT	STATUS	
reimbursement	Accommodation	₹22,222	in review	<button>View</button>
reimbursement	Meals	₹3,33,333	draft	<button>View</button>
reimbursement	Accommodation	₹1,850	draft	<button>View</button>
reimbursement	meals	₹4,500	in review	<button>View</button>
reimbursement	travel	₹16,600 (200.00 USD)	draft	<button>View</button>

GET /expenses/:id — get one (with approval chain)

Response 200

```
{
  "id": "uuid", "status": "in_review", "current_level": 1, "amount": "22000.00",
  "base_amount": "22000.00", "type": "reimbursement", "category": "travel",
  "steps": [
    { "id": "uuid", "level": 1, "required_role": "manager", "approver_id": "uuid", "status": "approved", "comment": "" },
    { "id": "uuid", "level": 2, "required_role": "finance", "approver_id": "uuid", "status": "pending", "comment": "" }
  ]
}
```

Errors: 404 not found / not in your tenant · 403 not allowed to view.

PATCH /expenses/:id — edit

Only the requester, only while `draft` / `submitted` / `in_review`. If the amount crosses a policy threshold while `in_review`, the approval chain is **rebuilt**. **Request** (any subset): `{ "amount": 20000, "currency": "INR", "category": "...", "description": "..." }` **Response** 200 updated expense. **Errors:** 403, 409, 422.

DELETE /expenses/:id

Requester only, drafts only. **Response** 200 → `{ "deleted": true }`. **Errors:** 403, 409.

POST /expenses/:id/submit

Moves `draft` → `in_review`, snapshots the policy, builds the `ApprovalStep` chain, and notifies **only the first approver**. Approvals are **sequential**: the next level's approver is notified only after the current level approves; a rejection ends the chain. Before building the chain the engine verifies an **eligible approver exists for every level** (excluding the requester) and fails fast otherwise. **Response 200** → expense with `status:"in_review"`, `current_level:1`. **Errors:** 403 not requester · 409 not a draft · 422 budget exceeded / no active policy / **no eligible approver for a required level**.

Budget-exceeded body:

```
{ "error": { "code": "UNPROCESSABLE", "message": "Budget exceeded for monthly (limit 2000000, already spent 1990000)", "details": { "ok": false, "period": "monthly", "limit": 2000000, "spent": 1990000 } } }
```

POST /expenses/:id/approve

Permission: `expense:approve`. Must be the assigned approver for the current level; cannot approve your own (approver resolution already routes an expense to a *different* eligible user, so it never lands in the requester's own queue). Approving the last level marks the expense `approved`; otherwise it advances `current_level` and notifies the next approver. Honors `Idempotency-Key`. **Request:** `{ "reason": "looks good" }` (optional). **Response 200** → expense; becomes `approved` after the last level, else stays `in_review` with `current_level` advanced. **Errors:** 403 wrong approver / self-approval · 404 · 409 not awaiting approval.

POST /expenses/:id/reject

Permission: `expense:approve`. **Reason is mandatory**. Terminates the chain. **Request:** `{ "reason": "not a business expense" }` **Response 200** → expense `status:"rejected"`. **Errors:** 400 no reason · 403 · 404 · 409.

POST /expenses/:id/withdraw

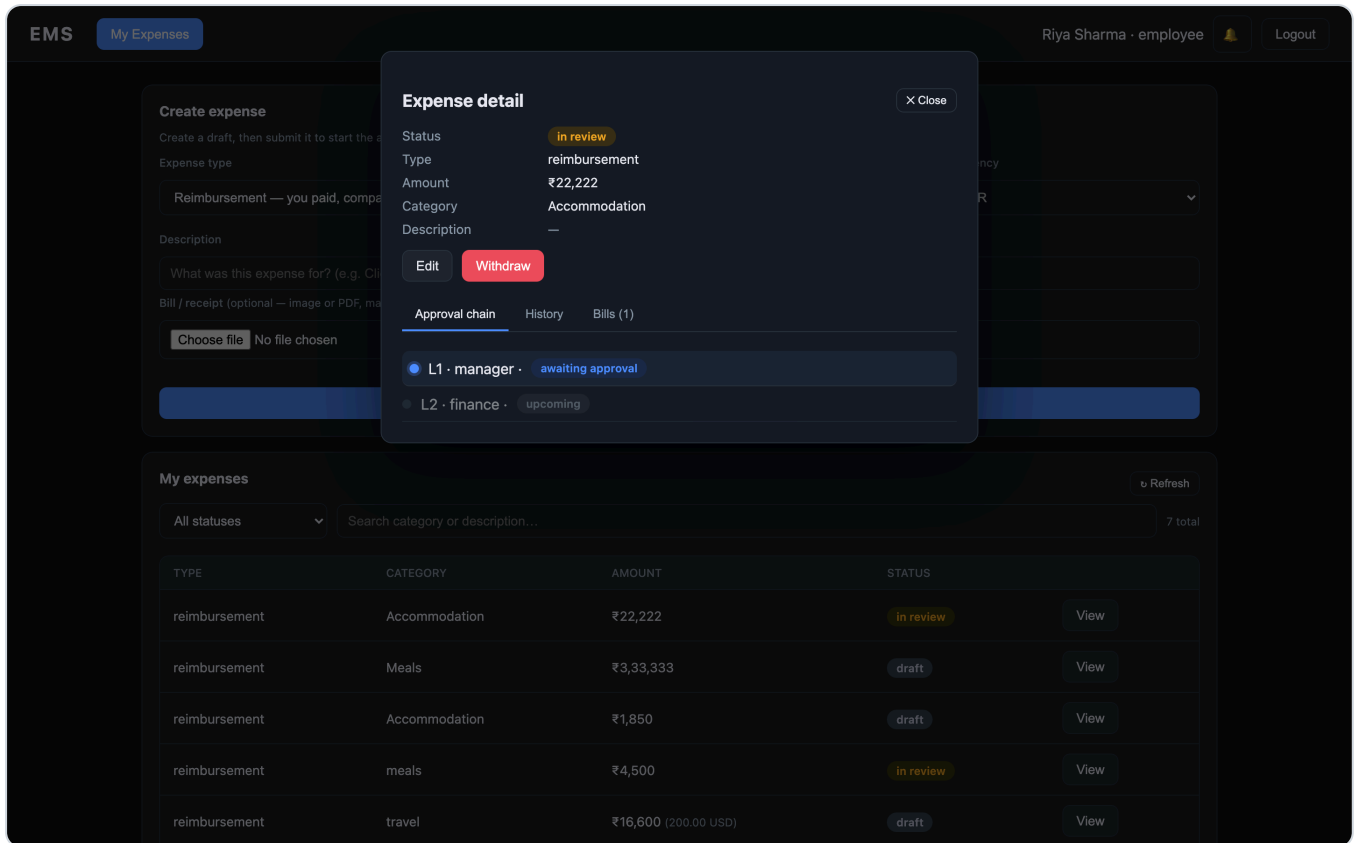
Requester cancels (`draft` / `submitted` / `in_review`). **Response 200** → `status:"withdrawn"`.

GET /expenses/:id/history — audit trail

Response 200

```
[
  {
    "id": "uuid",
    "actor_id": "uuid",
    "action": "expense.created",
    "entity_type": "expense",
    "entity_id": "uuid",
    "after_json": {
      "action": "expense.submitted",
      "after_json": {
        "status": "in_review",
        "levels": ["manager", "finance"]
      }
    },
    "reason": "ok",
    "created_at": "2026-..."
  },
  {
    "action": "expense.step_approved",
    "reason": "ok",
    "created_at": "2026-..."
  },
  {
    "action": "expense.approved",
    "created_at": "2026-..."
  }
]
```

UI — Expense detail (chain + audit trail): each expense shows its computed multi-level **approval chain** (level, role, status) and the immutable **history (audit trail)**. Requester actions (Submit/Withdraw) appear contextually.



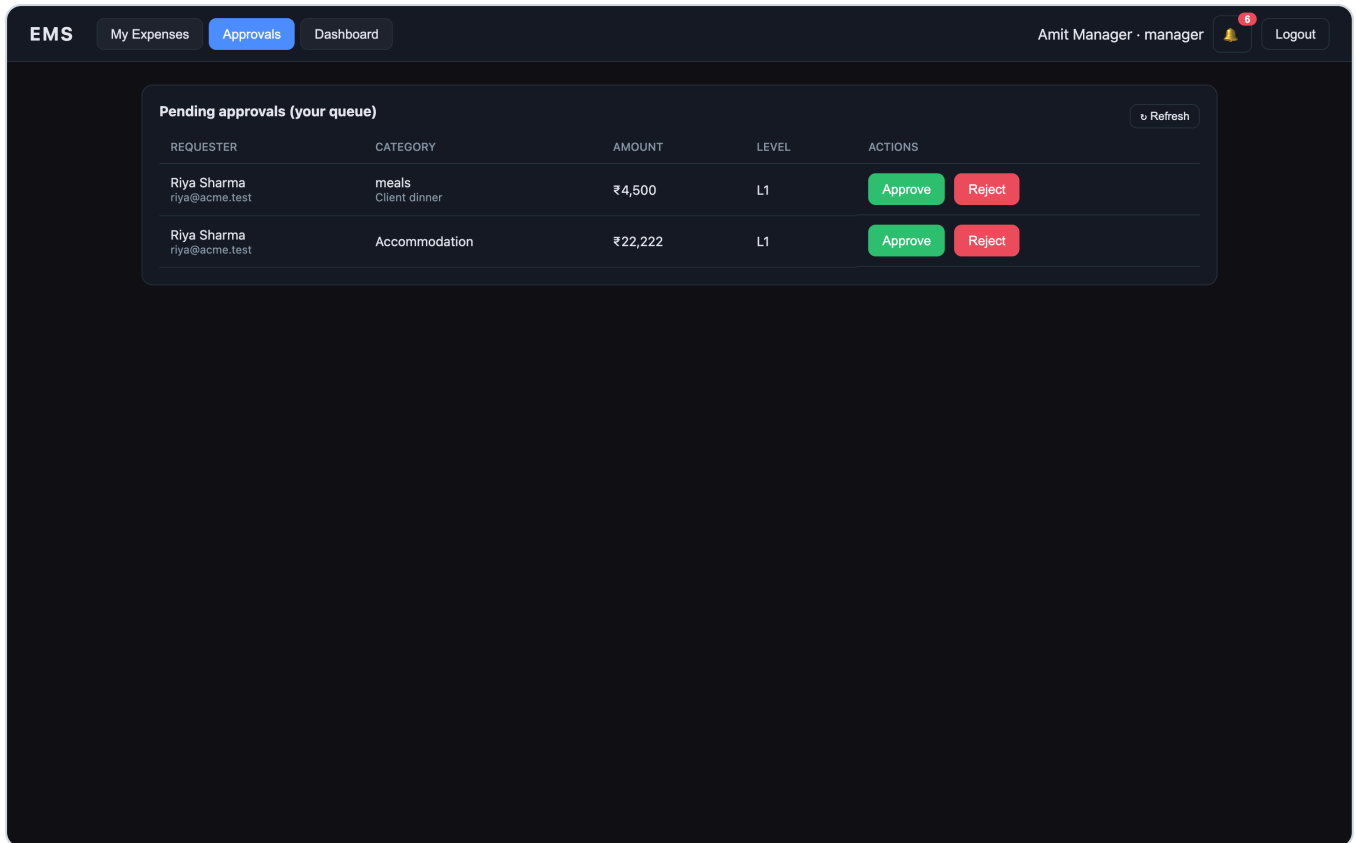
5. Approvals

GET /approvals/pending — my approval queue

Permission: `expense:approve` . Returns steps assigned to you that are pending at the expense's current level. **Response** 200

```
[
  {
    "id": "step-uuid",
    "expense_id": "uuid",
    "level": 1,
    "required_role": "manager",
    "status": "pending",
    "amount": "4500.00",
    "base_amount": "4500.00",
    "currency": "INR",
    "type": "reimbursement",
    "category": "meals",
    "description": "Client dinner",
    "requester_id": "uuid",
    "requester_name": "Riya Sharma",
    "requester_email": "riya@acme.test",
    "expense_status": "in_review"
  }
]
```

UI — Manager approvals queue: a manager sees only the steps assigned to them at the current level, and can Approve or Reject (reason prompted) directly.



6. Attachments (bill uploads)

Attach a **bill** — a receipt or invoice — to an expense. Files are uploaded to the API as multipart form data, the **bytes are stored through a `BlobStorage` abstraction** (default `LocalDiskStorage` ; production swaps in S3 with the same interface), and the **metadata** (filename, content-type, size, uploader) is written to Postgres. Allowed types: **PNG, JPEG, WebP, GIF, or PDF**, up to **5 MB**.

POST `/expenses/:id/attachments` — upload a bill

`multipart/form-data` with a single field `file` . Only the **requester** can attach a bill, and only while the expense is still editable (`draft` / `submitted` / `in_review`).

```
curl -X POST localhost:4000/expenses/$EID/attachments \
  -H "Authorization: Bearer $TOKEN" \
  -F "file=@/path/to/receipt.png;type=image/png"
```

Response 201

```
{
  "id": "uuid", "org_id": "uuid", "expense_id": "uuid",
  "s3_key": "ems-attachments/<org>/<expense>/<uuid>-receipt.png",
  "filename": "receipt.png", "content_type": "image/png",
  "size": 45512, "uploaded_by": "uuid", "uploaded_at": "2026-..."
}
```

Errors: `400` no file · `403` not the requester · `409` expense no longer editable · `422` unsupported file type / file too large.

GET /expenses/:id/attachments — list bills

Anyone allowed to view the expense (requester, their manager, finance/admin). **Response 200** → array of attachment metadata (no bytes).

GET /attachments/:id — download / preview a bill

Streams the stored bytes with the original `Content-Type` and an inline `Content-Disposition`. **Every download is authenticated and tenant-scoped** — bills are never served as public static files. Same visibility rules as the expense. **Errors:** `403` not allowed · `404` not found / other tenant.

UI — attach a bill at creation: the bill picker is on the **create expense** form, so a receipt/invoice can be attached up front (the UI creates the draft and uploads the bill in one step). It can also be added later from the detail **Bills** tab.

EMS

My Expenses

Riya Sharma · employee

Logout

Create expense

Create a draft, then submit it to start the approval flow.

Expense type

Reimbursement — you paid, company

Category

Accommodation

Amount

1850

Currency

INR

Description

Client dinner — receipt attached

Bill / receipt (optional — image or PDF, max 5MB)

Choose file ems-sample-receipt.png

Create draft

My expenses

Refresh

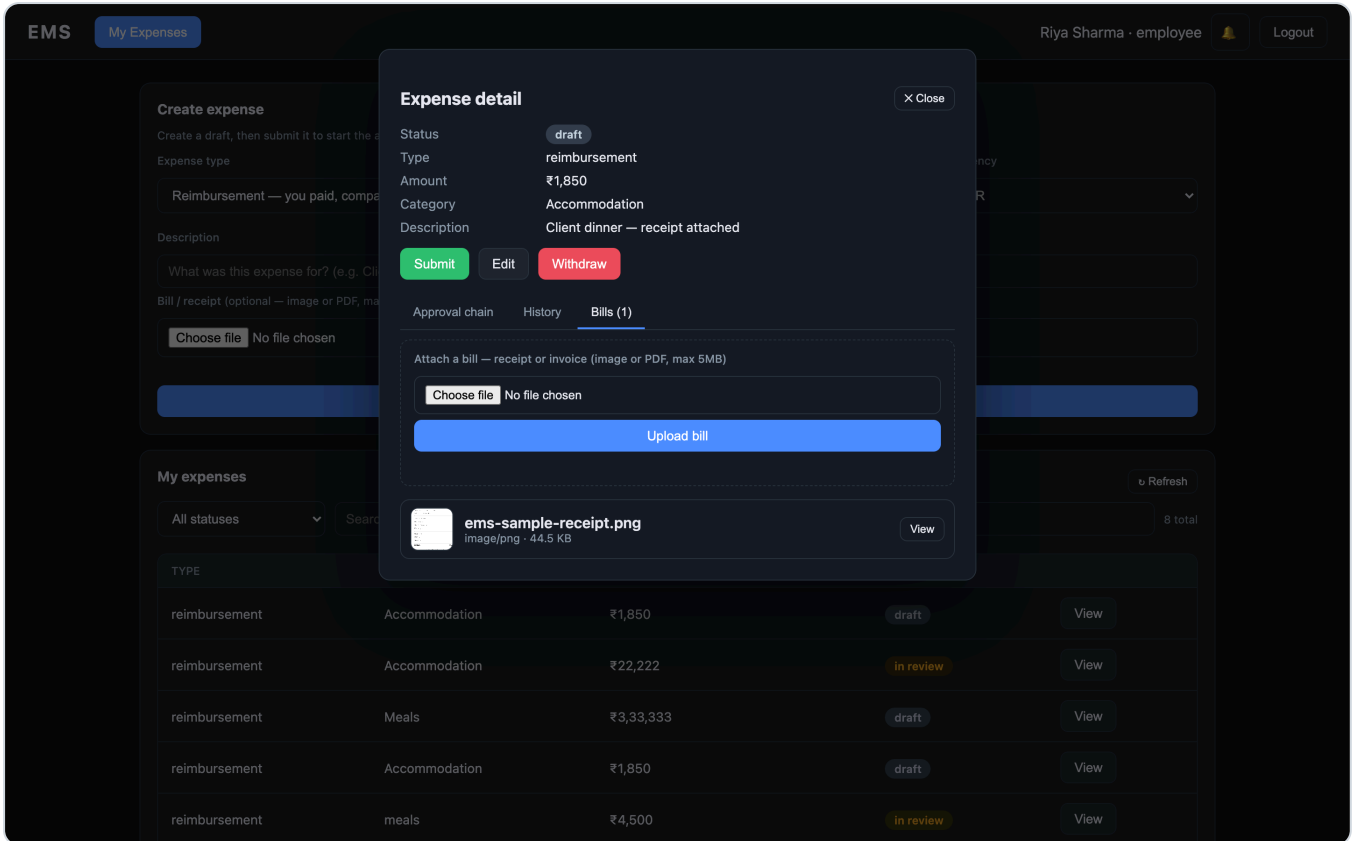
All statuses

Search category or description...

7 total

TYPE	CATEGORY	AMOUNT	STATUS	
reimbursement	Accommodation	₹22,222	in review	View
reimbursement	Meals	₹3,33,333	draft	View
reimbursement	Accommodation	₹1,850	draft	View
reimbursement	meals	₹4,500	in review	View
reimbursement	travel	₹16,600 (200.00 USD)	draft	View

UI — Bills tab: uploaded bills show as thumbnails (images) and open in a new tab via **View**. Reviewers see the same bills while approving.



POST /attachments/presign — mock S3 presigned URL (documented prod path)

The alternative to API-proxied uploads is a **direct-to-bucket** PUT using a pre-signed URL. This endpoint returns a *mock* URL to document that flow; no real S3 call is made in the prototype. **Request:** `{ "filename": "bill.png", "contentType": "image/png" }` **Response 200**

```
{
  "key": "ems-attachments/<org>/<uuid>-bill.png",
  "uploadUrl": "https://mock-s3.local/...?X-Amz-Signature=mock&expires=900",
  "method": "PUT", "expiresInSeconds": 900,
  "note": "MOCK presigned URL (no real S3 in prototype)"
}
```

Errors: 400 missing filename/contentType.

7. Policies

GET /policies

Permission: analytics:view . **Response 200** → array:

```
[{ "id":"uuid","name":"Default approval policy","tolerance_percent":"10.00","active":true,"version":1,
  "rules_json":{ "currency":"INR","rules":[
    { "min":0,"max":5000,"levels":["manager"]},
    { "min":5001,"max":50000,"levels":["manager","finance"]},
    { "min":50001,"max":null,"levels":["manager","finance","admin"]}]]}]
```

Exactly one policy is active per org. The approval engine always uses the single active policy, so the UI never shows several "active" rows competing to route an expense.

POST /policies — create

Permission: `policy:manage` . Creating a policy as `active` (the default) **deactivates every other policy in the org in the same transaction**. Policy names must be unique within the org. **Request**

```
{ "name": "Travel policy", "tolerancePercent": 10,
  "rulesJson": { "currency": "INR", "rules": [ { "min": 0, "max": 10000, "levels": ["manager"] } ] } }
```

Response `201` policy object. **Errors:** `400` invalid · `409` duplicate policy name · `422` a rule's `max < min` .

PATCH /policies/:id — update (bumps version)

Any subset of `name` , `tolerancePercent` , `active` , `rulesJson` . Setting `active:true` activates this policy and **deactivates all others** atomically. **Errors:** `404` if missing · `409` duplicate name.

DELETE /policies/:id

Response `200` → `{ "deleted": true }` . **Errors:** `404` if missing · `400` the **active policy can't be deleted** (activate another one first).

UI — Policies: a visual rule builder (amount band → ordered approver chain, no JSON to hand-write) plus the tolerance percentage. The list shows which policy is active, lets you activate/deactivate with one click, and delete inactive ones. Admins also manage the org's **expense categories** here (the list that feeds the expense-form dropdown).

The screenshot shows the 'Policies' tab in the EMS application. The header includes 'EMS', navigation links for 'My Expenses', 'Approvals', 'Dashboard', and 'Policies', and user information 'Neha Finance · finance' with a notification bell and a 'Logout' button. The main content area is titled 'Create approval policy' and includes a sub-header: 'Only one policy is active at a time — the active policy decides who approves each expense based on its amount. Creating a policy makes it the active one.'

The form consists of several sections:

- Policy name:** A text input field with the placeholder 'e.g. Standard approval policy'.
- Tolerance % (allowed variance for company-paid amounts):** A numeric input field with the value '0'.
- Approval bands — for each amount range, tick who must approve:** A section with three rows, each representing an amount band. Each row has two input fields for 'From (₹)' and 'To (₹, blank = no limit)', and a list of checkboxes for 'Must be approved by (in this order)'.
 - Row 1: From (₹) is '0', To (₹) is '5000'. Approved by: ☒ manager, ☐ finance, ☐ admin.
 - Row 2: From (₹) is '5001', To (₹) is '50000'. Approved by: ☒ manager, ☒ finance, ☐ admin.
 - Row 3: From (₹) is '50001', To (₹) is '∞'. Approved by: ☒ manager, ☒ finance, ☒ admin.
- + Add band** and **Use standard template** buttons at the bottom.

7b. Categories

Per-org expense categories that populate the category dropdown on the expense form. Seeded with sensible defaults on signup; admins can add/remove more.

GET /categories

Permission: any authenticated user. **Response** 200

```
[{ "id":"uuid","org_id":"uuid","name":"travel","active":true,"created_at":"2026-..." }]
```

POST /categories — add a category

Permission: policy:manage . **Request:** { "name": "software" } **Response** 201 category object. **Errors:** 400 invalid · 409 duplicate name.

DELETE /categories/:id — soft-deactivate

Permission: policy:manage . Marks the category inactive (history on existing expenses is preserved). **Response** 200 → { "deleted": true } . 404 if missing.

8. Budgets

GET /budgets (perm analytics:view) → array of budgets.

POST /budgets (perm budget:manage)

Request: { "userId": "uuid|null", "scope": "user|org", "period": "daily|monthly", "limitAmount": 50000, "currency": "INR" } **Response** 201 budget object. **Errors:** 400 invalid.

GET /budgets/utilization — current user's usage

Response 200 → { "monthly": { "limit": 2000000, "spent": 25200, "pct": 1 } }

9. Users

GET /users (perm user:manage) → array (password hashes stripped).

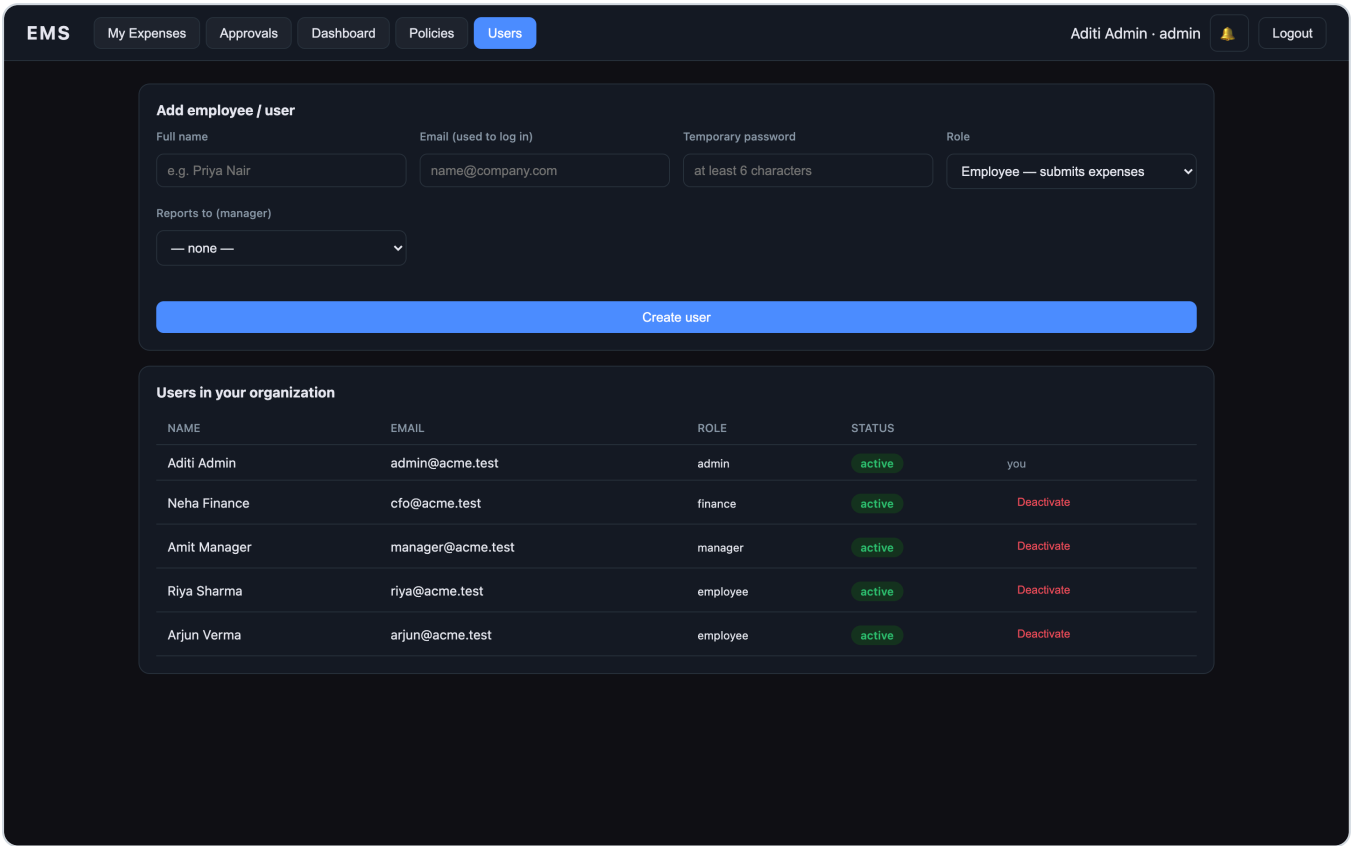
POST /users (perm user:manage)

Request: { "name":"New", "email":"new@acme.test", "password":"secret1", "role":"employee", "managerId":"uuid|null" } **Response** 201 user (no hash). **Errors:** 400 invalid · 409 duplicate email.

PATCH /users/:id (perm user:manage)

Any subset of `name` , `role` , `managerId` , `isActive` . Setting `isActive:false` **deactivates** the user — they can no longer log in and are skipped by approver resolution; `isActive:true` reactivates them. **Guards:** you cannot deactivate **your own account**, nor the **last active admin** in the org (→ `400` / `409`). `404` if missing.

UI — Admin Users: admins add employees/managers/finance to their org (with optional manager assignment), view the org roster with active/inactive status, and deactivate or reactivate users from the table.



10. Notifications

`GET /notifications` → my notifications (newest 100).

```
{ { "id":"uuid","type":"approval_requested","payload_json":{"expenseId":"uuid","level":1},"read":false
```

`POST /notifications/:id/read` → `{ "read": true }` . `404` if not yours.

11. Analytics / Observability

All require `analytics:view` . Summary is Redis/memory-cached (short TTL).

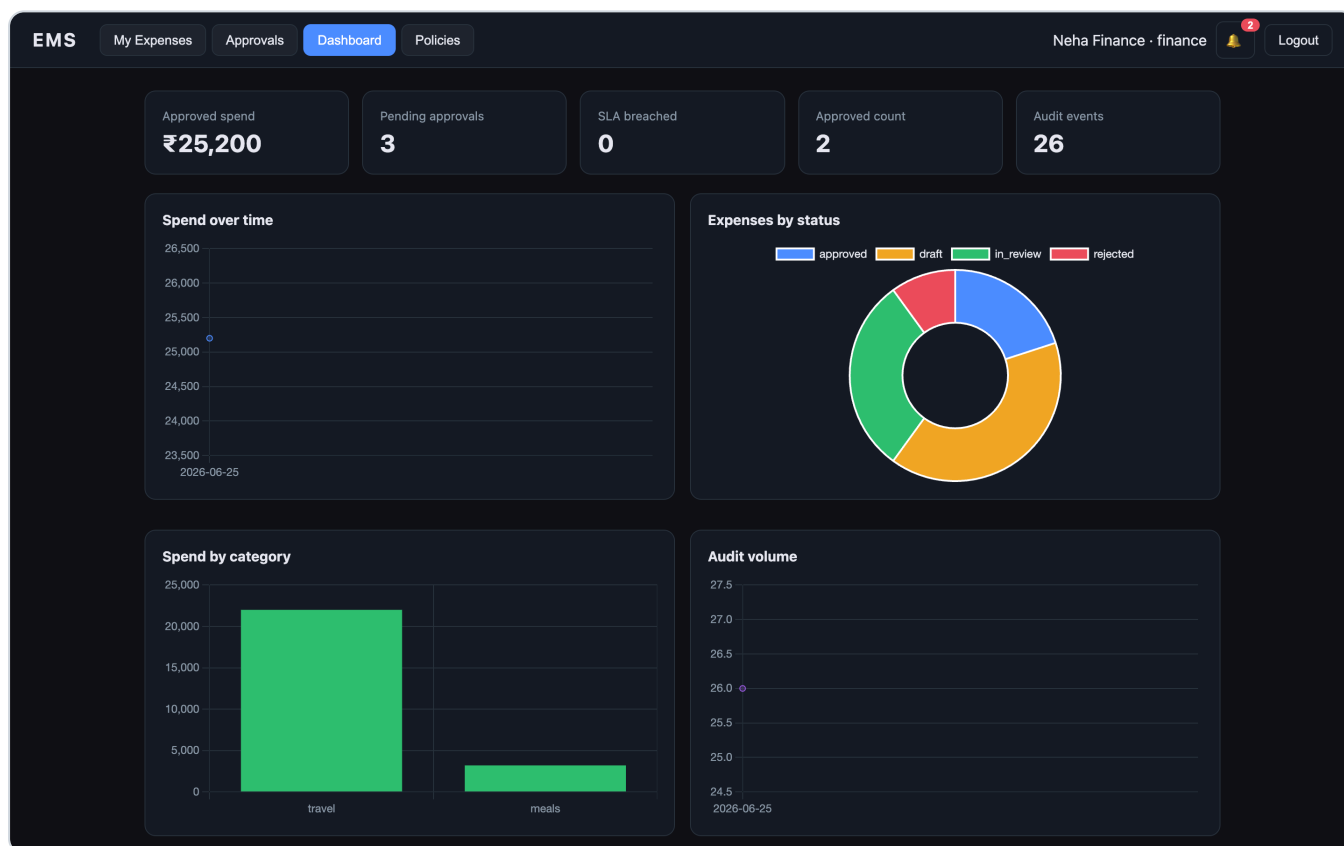
Endpoint	Response
<code>GET /analytics/summary</code>	<code>{ approvedSpend, approvedCount, pendingApprovals, slaBreached, auditEvents, byStatus:{...} }</code>

Endpoint	Response
GET /analytics/pend	[{ "day":"2026-06-24", "total": 25200 }]
GET /analytics/by-status	[{ "status":"approved", "count": 2 }]
GET /analytics/by-category	[{ "category":"travel", "total": 22000 }]
GET /analytics/audit-volume	[{ "day":"2026-06-24", "count": 16 }]

Example GET /analytics/summary :

```
{ "approvedSpend": 25200, "approvedCount": 2, "pendingApprovals": 2, "slaBreached": 0,
  "auditEvents": 16, "byStatus": { "approved": 2, "draft": 1, "in_review": 2, "rejected": 1 } }
```

UI — Finance/Admin observability dashboard: stat cards (approved spend, pending, SLA breached, audit events) plus charts — spend over time, expenses by status, spend by category, audit volume. Data is aggregated from Postgres and cached.



12. Security, isolation & concurrency

Authentication. Access tokens are short-lived JWTs ({ id, org_id, role, email }). Refresh tokens are **opaque random strings stored only as SHA-256 hashes** in `refresh_tokens` . `/auth/refresh` does **single-use rotation** (old token revoked, new one issued, chained via `replaced_by`); replaying a revoked token is treated as theft and revokes the user's whole session family. `/auth/logout` revokes a token. (Code: `src/auth/refreshToken.ts` .)

Tenant isolation. `org_id` is read from the verified token (never the client) and applied to every query — the core isolation guardrail. This is app-layer isolation; one missing predicate would leak across tenants. The production hardening is **Postgres Row-Level Security**:

```
ALTER TABLE expense_requests ENABLE ROW LEVEL SECURITY;
CREATE POLICY tenant_isolation ON expense_requests
  USING (org_id = current_setting('app.org_id')::uuid);
-- set per transaction: SET LOCAL app.org_id = '<org from token>';
```

It's documented rather than enabled here because RLS interacts with the connection pool and the migration/seed paths; the app-layer guard plus this plan is the deliberate tradeoff.

Concurrency (no double-apply). Every state transition (submit/edit/approve/reject/withdraw) opens a transaction and takes an exclusive row lock via `SELECT ... FOR UPDATE ( getByIdForUpdate )`. Concurrent approvals on the same expense are serialized by Postgres: the loser blocks, re-reads the committed state, and is rejected by the status/level guards. Idempotency keys are orthogonal — they dedupe identical retries, the row lock handles distinct concurrent requests.

13. End-to-end flow (recap)

```
signup/login → create expense (draft) → submit
  → engine reads policy, snapshots it, builds ApprovalStep chain by amount
  → notify approver(s)
manager approve → (advance level) → finance approve → ... → APPROVED
                |
                reject (reason) → REJECTED
every transition → immutable audit_log entry (in the same DB transaction)
finance/admin → dashboard reads aggregates for live observability
```